

Apply filters to SQL queries

Project description

In this project, I used SQL queries were used to investigate potential security incidents by analyzing login activity and employee data stored in an organizational database. The focus was on applying filters with WHERE, AND, OR, and NOT operators to retrieve specific records that may indicate suspicious behavior. The `log_in_attempts` table was queried to identify failed login attempts, after-hours access, activity on specific dates, and login attempts originating outside of Mexico. These queries help highlight potential unauthorized access or policy violations. The `employees` table was also queried to retrieve employee information based on department, such as identifying users in Marketing, Finance, or Sales, as well as employees not in the IT department.

You recently discovered a potential security incident that occurred after business hours. To investigate this, you need to query the **log_in_attempts** table and review after hours login activity. Use filters in SQL to create a query that identifies all failed login attempts that occurred after 18:00. (The time of the login attempt is found in the **login_time** column. The **success** column contains a value of **0** when a login attempt failed; you can use either a value of **0** or **FALSE** in your query to identify failed login attempts.)

Describe your query and how it works in the Retrieve after hours failed login attempts section of the Apply filters to SQL queries template.

Retrieve after hours failed login attempts

```
SELECT * FROM log_in_attempts WHERE success = 0 AND login_time > '18:00:00';
```

The SQL statement above identifies what attempts were failed and which failed attempts occurred after 18:00 (6 PM)

A suspicious event occurred on 2022-05-09. To investigate this event, you want to review all login attempts which occurred on this day and the day before. Use filters in SQL to create a query that identifies all login attempts that occurred on 2022-05-09 or 2022-05-08. (The date of the login attempt is found in the **login_date** column.)

Describe your query and how it works in the Retrieve login attempts on specific dates section of the Apply filters to SQL queries template.

Retrieve login attempts on specific dates

```
SELECT * FROM log_in_attempts WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

The SQL statement above identifies which login attempts happened on either days

There's been suspicious activity with login attempts, but the team has determined that this activity didn't originate in Mexico. Now, you need to investigate login attempts that occurred outside of Mexico. Use filters in SQL to create a query that identifies all login attempts that occurred outside of Mexico. (When referring to Mexico, the **country** column contains values of both **MEX** and **MEXICO**, and you need to use the **LIKE** keyword with % to make sure your query reflects this.)

Describe your query and how it works in the Retrieve login attempts on specific dates section of the Apply filters to SQL queries template.

Retrieve login attempts outside of Mexico

```
SELECT * FROM log_in_attempts WHERE NOT country LIKE "MEX%";
```

The SQL statement above identifies where login attempts happened anywhere but Mexico and also uses the LIKE operator with MEX% to retrieve any words starting with MEX such as MEX itself and MEXICO

Your team wants to perform security updates on specific employee machines in the Marketing department. You're responsible for getting information on these employee machines and will need to query the **employees** table. Use filters in SQL to create a query that identifies all employees in the Marketing department for all offices in the East building.

(The department of the employee is found in the **department** column, which contains values that include **Marketing**. The office is found in the office column. Some examples of values in this column are **East-170**, **East-320**, and **North-434**. You'll need to use the **LIKE** keyword with % to filter for the East building.)

Describe your query and how it works in the Retrieve employees in Marketing section of the Apply filters to SQL queries template.

Retrieve employees in Marketing

```
SELECT * FROM employees WHERE department = 'Marketing' AND office LIKE 'East%';
```

The SQL statement above identifies the employees in the marketing department and the office they are in starting with the word EAST such as EAST-120 or EAST-170 and so on

Your team now needs to perform a different security update on machines for employees in the Sales and Finance departments. Use filters in SQL to create a query that identifies all employees in the Sales or Finance departments. (The department of the employee is found in the **department** column, which contains values that include **Sales** and **Finance**.)

Describe your query and how it works in the Retrieve employees in Finance or Sales section of the Apply filters to SQL queries template.

Retrieve employees in Finance or Sales

```
SELECT * FROM employees WHERE department = 'Finance' OR department = 'Sales';
```

The SQL statement above identifies the employees in the finance or sales department

Your team needs to make one more update to employee machines. The employees who are in the Information Technology department already had this update, but employees in all other departments need it. Use filters in SQL to create a query which identifies all employees not in the IT department. (The department of the employee is found in the **department** column, which contains values that include **Information Technology**.)

Describe your query and how it works in the Retrieve all employees not in IT section of the Apply filters to SQL queries template.

Retrieve all employees not in IT

```
SELECT * FROM employees WHERE department != 'Information Technology';
```

The SQL statement above identifies any employees except those in the Information Technology department

Summary

In this activity, SQL filtering techniques were used to analyze login activity and employee data. The queries applied WHERE clauses with logical operators such as AND, OR, and != to narrow results based on time, date, location, and department. These filters help identify failed

after-hours login attempts, suspicious access from outside Mexico, and specific employee groups, supporting effective security investigation and access management